

Question Bank

Unit II: Introduction to Industrial Control Systems (09 Hrs.)

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Syllabus: Implementation of digital controller; control strategies for industrial processes; programmable logic controller; real-time issues on signal transmission and control; communication systems for industrial automation; data acquisition and supervisory control; control of discrete manufacturing processes; intelligent systems for monitoring, supervision and control; case studies of industrial control systems.

Part A 10 multiple-choice questions $10 \times 1 = 10$ marks

Part B 5 short-answer questions $5 \times 2 = 10$ marks

Part C 3 long-answer questions $3 \times 10 = 30$ marks

Part A: Multiple-Choice Questions

1 mark each

Q1. A 12-bit ADC divides its input range into how many levels?

- (a) 256
- (b) 1024
- (c) 4096
- (d) 65 536

Answer: (c) $2^{12} = 4096$ levels.

Q2. According to the Nyquist criterion, a signal must be sampled at least

- (a) twice its highest frequency
- (b) at its own frequency
- (c) half its highest frequency
- (d) once per second

Answer: (a) Twice its highest frequency ($f_s > 2f_{\max}$); otherwise aliasing occurs.

Q3. Which of the following is **not** an IEC 61131-3 PLC programming language?

- (a) Ladder Diagram
- (b) HTML
- (c) Function Block Diagram
- (d) Structured Text

Answer: (b) HTML.

Q4. The correct order of a PLC scan cycle is

- (a) execute → read → write
- (b) read → write → execute
- (c) write → read → execute
- (d) read inputs → execute program → update outputs

Answer: (d) Read inputs → execute program → update outputs (then housekeeping).

Q5. SCADA stands for

- (a) Supervisory Control And Data Acquisition
- (b) System Control And Data Analysis
- (c) Sequential Control And Digital Automation
- (d) Safety Control And Data Alarm

Answer: (a) Supervisory Control And Data Acquisition.

Q6. In cascade control, the inner (slave) loop should be

- (a) slower than the outer loop (b) the same speed as the outer loop
(c) open loop (d) faster than the outer loop

Answer: (d) Faster — typically 3 to 10 times faster than the outer loop.

Q7. A hard real-time system is one in which

- (a) only the average speed matters (b) deadlines do not matter
(c) a missed deadline is a failure (d) tasks run once a day

Answer: (c) Missing a deadline is a system failure (e.g. emergency shutdown, airbag).

Q8. Modbus RTU uses which method of medium access?

- (a) Token ring (b) Master–slave polling
(c) CSMA/CD (d) Time-slot broadcast

Answer: (b) Master–slave polling.

Q9. NMEA 2000, used on ships for navigation and engine data, is based on

- (a) RS-232 (b) Wi-Fi
(c) Profibus (d) CAN bus

Answer: (d) CAN bus.

Q10. In a Sequential Function Chart (SFC), a step drawn with a double border is the

- (a) final step (b) alarm step
(c) initial step (d) transition

Answer: (c) Initial step — active when the sequence starts.

Part B: Short-Answer Questions

2 marks each

Q1. Find the resolution of a 12-bit ADC with an input range of 0–10 V.

[2]

Answer

$q = \frac{\text{FSR}}{2^n} = \frac{10}{2^{12}} = \frac{10}{4096} = 2.44 \text{ mV}$. A change smaller than 2.44 mV cannot be seen by the controller.

Q2. Write the velocity (incremental) form of the discrete PID algorithm and state one advantage. [2]

Answer

$\Delta u[k] = K_p (e[k] - e[k-1]) + K_i T e[k] + \frac{K_d}{T} (e[k] - 2e[k-1] + e[k-2])$, $u[k] = u[k-1] + \Delta u[k]$.
Advantage: no integral windup (only the output needs clamping) and bumpless manual/auto transfer.

Q3. What is feedforward control? Give one example.

[2]

Answer

Feedforward control **measures a disturbance** and corrects the manipulated variable **before** the disturbance affects the process variable. It is combined with feedback to trim remaining errors. Example: in boiler drum-level control, the steam flow is measured and the feedwater is increased at once when steam demand rises.

Q4. Define latency and jitter in a real-time control system.

[2]

Answer

Latency is the delay between an event (or sampling instant) and the system's response to it. **Jitter** is the variation of that delay from one cycle to the next. Both reduce the phase margin of a control loop; jitter also makes the sample time irregular.

Q5. Give any two differences between a PLC and a DCS.

[2]

Answer

- (i) A PLC is mainly used for fast **logic and machine control** of one machine or cell; a DCS is used for **continuous process control** with hundreds of PID loops across a plant.
- (ii) A PLC system is usually smaller and cheaper, programmed in ladder/ST; a DCS is an integrated vendor system with its own high-reliability network, operator stations and historian.

Part C: Long-Answer Questions

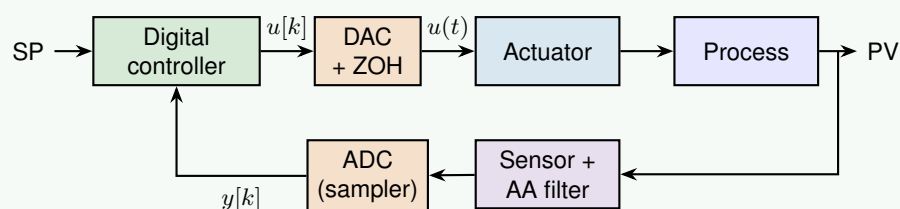
10 marks each

Q1. Draw the block diagram of a digital (computer) control loop and explain each element. Discuss sampling, quantisation and the choice of sample time. Give a flowchart for implementing a PID algorithm on a microcontroller.

[10]

Answer

(a) Block diagram



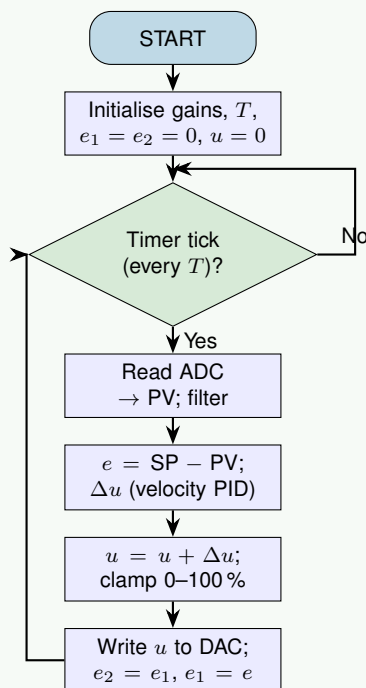
- **Sensor + anti-aliasing (AA) filter** — measures PV and removes frequencies above $f_s/2$.
- **ADC (sampler)** — converts PV to a number $y[k]$ every T seconds.
- **Digital controller** — a CPU running the control algorithm (filter, PID, limits).
- **DAC + zero-order hold** — converts $u[k]$ to an analog signal and holds it constant until the next sample ("staircase" output).
- **Actuator & process** — as in any control loop.

(b) Sampling and quantisation

- **Sampling:** the controller sees the process only at $t = kT$. Sample at $f_s > 2f_{\max}$ (Nyquist) to avoid **aliasing**.
- **Quantisation:** an n -bit ADC has step $q = \text{FSR}/2^n$ (12-bit, 0–10 V $\Rightarrow$ 2.44 mV); finer resolution needs more bits.

- **Choice of T :** $T \leq \tau/10$ (τ = dominant time constant) or 10–20 samples per rise time. Too large T adds delay and can destabilise the loop; too small T wastes CPU time and amplifies noise in the D term.

(c) Implementation flowchart



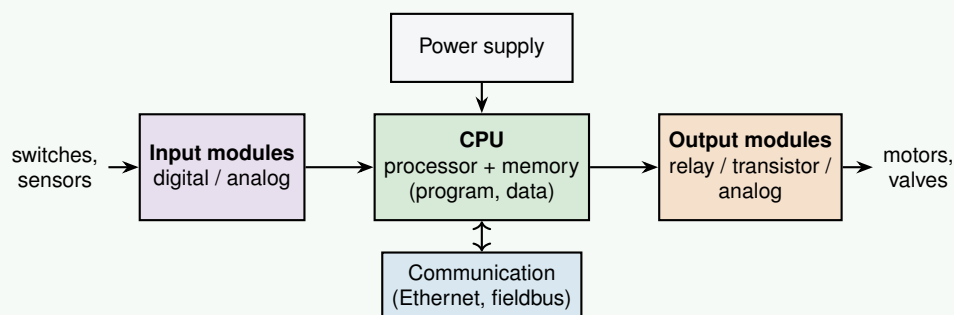
Explanation

- A **hardware timer interrupt** starts each cycle so that T is constant (no jitter from program length).
- The measurement is read and **filtered**: $y_f[k] = a y_f[k-1] + (1-a)x[k]$.
- The **velocity-form PID** computes only the change Δu ; clamping u provides anti-windup.
- The output is written, past errors are shifted, and the controller waits for the next tick.
- A **watchdog** resets the controller if a cycle is not completed in time.

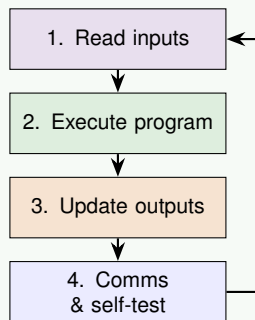
Q2. With a block diagram, explain the architecture of a Programmable Logic Controller. Explain the PLC scan cycle and its response time. Draw a ladder diagram for a motor start/stop circuit with seal-in. **[10]**

Answer

(a) PLC architecture



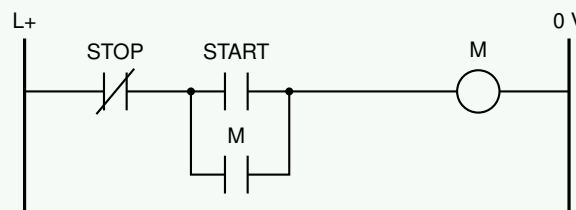
- **Input modules** convert field signals (24 V DC, 4–20 mA) to logic levels, with isolation and filtering.
- **CPU** executes the user program stored in memory, holds the input/output image tables, timers and counters.
- **Output modules** drive contactors, solenoid valves, lamps and analog actuators.
- **Power supply** and **communication** modules (programming PC, HMI, SCADA, remote I/O).

(b) Scan cycle**Response time**

The scan repeats every 1–50 ms. An input that changes just after being read is only seen in the next scan, so the worst case is

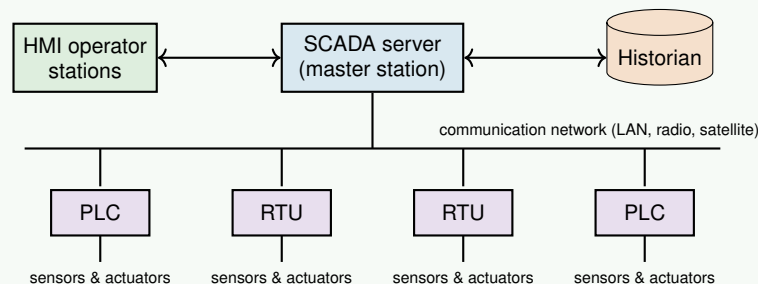
$$t_{\text{resp}} \approx t_{\text{in}} + 2t_{\text{scan}} + t_{\text{out}}$$

e.g. input filter 3 ms, scan 5 ms, relay output 10 ms
 $\Rightarrow 3 + 10 + 10 = \mathbf{23 \text{ ms.}}$

(c) Motor start/stop ladder diagram

Pressing **START** energises coil **M**; the parallel **M** contact *seals in* so the motor keeps running after **START** is released. Pressing **STOP** (normally closed) breaks the rung and **M** drops out. After a power failure **M** stays off (no automatic restart). Equivalent Boolean expression:
 $M = (\text{START} + M) \cdot \text{STOP}.$

Q3. Explain the architecture and functions of a SCADA system with a neat diagram. Compare SCADA with DCS and PLC systems, and list good practices for alarm management. **[10]**

Answer**(a) SCADA architecture**

- **Field devices** — sensors and actuators on the process.
- **RTUs / PLCs** — collect field data, run local control, send data on request.
- **Communication network** — links remote sites to the master (Ethernet, radio, GSM, satellite).
- **Master station** — polls the RTUs, processes data, runs supervisory logic.
- **HMI** — mimic diagrams, alarms, trends, operator commands.
- **Historian** — long-term database for reports and analysis.

Functions: data acquisition, monitoring, supervisory control (set points, start/stop), alarm handling, trending, reporting, event logging.

(b) Comparison

Feature	PLC	DCS	SCADA
Main job	fast logic, machines	continuous process control	supervision over wide areas
Area	one machine / cell	one plant	pipelines, grids, fleets
Control loops	some PID	hundreds of PID loops	mostly in RTUs / PLCs
Example	packaging line	refinery, power plant	water network

(c) Alarm management good practice (ISA-18.2 / EEMUA 191)

- Every alarm must require a defined **operator action**; otherwise make it an event, not an alarm.
- **Prioritise** alarms (critical, high, medium, low) and show priority by colour/sound.
- Avoid **alarm floods**: target less than about one alarm per 10 minutes in normal operation.
- Use **dead-bands** and on/off delays to stop chattering alarms; suppress consequential alarms.
- Review alarm statistics regularly and remove nuisance alarms.